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A customized MobileNetV3Large-based deep learning framework for plant disease detection

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Abstract

A wide range of plant diseases that impact economically and nutritionally significant crops such as Beans, Cauliflower, Jackfruit, Malabar Spinach, and Mango pose a serious danger to Bangladesh's agricultural output. Scalable, quick, and precise solutions are needed because traditional disease inspection techniques are lengthy and frequently out of reach for rural farmers. Based on a modified MobileNetV3Large architecture, this study suggests a lightweight, portable, and effective framework for detecting plant diseases. The model was trained on a hybrid dataset consisting of 26 classes that combined field-collected images from Bangladesh with publicly available sources, utilizing transfer learning and fine-tuning techniques to ensure robustness against real-world conditions such as changing lighting and background clutter. In order to improve model generalization and reduce overfitting, data augmentation techniques such as random flipping, rotation, brightness and contrast change, and zooming were utilized. Under identical experimental conditions, the performance of the suggested model was methodically compared to a customized Convolutional Neural Network, baseline MobileNetV3Large, and EfficientNetB1 utilizing metrics such as test accuracy, confusion matrix, precision, recall, F1-score, ROC-AUC, and precision-recall curves. Compared with the baseline MobileNetV3Large (98.97%), EfficientNetB1 (98.3%), and the custom Convolutional Neural Network (95.51%), the customized MobileNetV3Large demonstrated better predictive capability and generalizability across all classes, achieving superior test accuracy (99.37%). The outcomes confirm the system's excellent diagnostic accuracy and viability for immediate use in mobile-based digital agriculture systems in rural regions with limited resources.

Keywords Convolutional neural network (CNN), Deep learning, MobileNetV3, Plant disease, Transfer learning

1 Introduction

The economy of Bangladesh and its food security are largely dependent on agriculture. This sector supports the livelihoods of millions of farmers, home gardeners and rural households who rely on local harvests for daily sustenance and revenue [1]. However, a portion of agricultural products are seriously threatened by leaf diseases, which can lead to lower yields, monetary losses, and even risks to the country's food stability [2]. Numerous types of crops are grown in Bangladesh, among them five species, beans,

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cauliflower, jackfruit, malabar spinach, and mango stand out due to their high nutritional content, significant economic impact, and appeal to both large and small-scale growers. Therefore, protecting these crops from disease is crucial to sustain productivity and promote dietary security for all.

Leaf diseases, often caused by bacterial, viral, or fungal pathogens, can be challenging to detect using conventional inspection techniques because of overlapping symptoms, subtle early-stage features, and the requirement for specialized knowledge [3]. In rural areas of Bangladesh, where access to trained agronomists or laboratory facilities are limited, these challenges are even more severe [4]. Inaccurate or delayed disease diagnosis can have serious consequences, including uncontrolled spread, overuse of pesticides, and long-term damage to the soil and surrounding ecosystems.

Efficient disease detection systems could also support urban or suburban residents who maintain small home gardens. These gardens enhance local food security, promote nutritional diversity, and supply fresh produce to households [5]. As home growers typically lack specialized expertise to identify diseases, an accessible, automated detection tool would greatly help them protect their modest harvests.

Convolutional Neural Networks (CNNs) have emerged as a powerful tool for automating image-based disease diagnosis in deep learning applications [6, 7]. Lightweight CNN architectures such as MobileNetV3 are particularly well-suited for real-time or mobile-based applications in resource-constrained settings because they provide high classification accuracy with minimal computational complexity [8]. Transfer learning techniques further enhance these models by adapting pre-trained feature extractors to agricultural datasets with fewer labeled samples [9]. However, many current studies rely on ideal datasets collected in controlled laboratory environments, which do not accurately reflect the variety of backgrounds, lighting conditions, and occlusions present in field situations [10].

This study fills these gaps by developing a customized deep learning framework based on MobileNetV3Large, trained on a hybrid dataset that includes both the field-collected and publicly available images of different crops grown in Bangladesh. Advanced data augmentation techniques are used to improve the proposed model's generalization in practical scenarios. The framework aims to achieve high accuracy while maintaining computational efficiency by comparing its performance to baseline MobileNetV3, EfficientNetB1, and a custom CNN. This could allow deployment on low-cost mobile devices for both home gardeners and farmers.

2 Literature study

The need for quick and precise diagnosis in agriculture has encouraged research into deep learning-based plant disease detection. Because of their automatic feature learning capabilities, CNNs in particular have shown great efficacy for image-based plant disease classification.

E. Elfatimi et al. [11] suggested a MobileNet-based framework for beans that targets both healthy leaves and two disease classes. They experimented with MobileNetV1 and MobileNetV2 architectures and gathered 1296 field photos, attaining a 92.97% test accuracy. In a similar vein, Using deep learning models such as GoogleNet and VGG16, which were trained on 1296 leaf images from three classes (Angular Leaf Spot, Bean Rust, and Healthy), Sahu et al. [12] enhanced interpretability in bean disease classification. They

used visualization techniques like CNN filter visualizations, heat maps, and intermediate layer activations to locate diseased regions and make the model's decision-making process understandable to non-experts. They used GoogleNet to achieve the highest accuracy of 95.31%. Singh et al. [13] used MobileNetV2, EfficientNetB6, and NasNet to apply transfer learning on a dataset of 1295 images, 3 classes of bean leaves. They reported that EfficientNetB6 had the best accuracy of 91.74%, and they demonstrated how optimizers affect CNN performance for early disease detection in agriculture.

Haque and Mojumdar [14] developed a CNN-based classification model for detecting scab disease on Malabar leaves, achieving a notable accuracy of 96.77% using RGB images resized to 200×200, surpassing prior works which reached up to 85%.

Using a dataset of 1275 high-resolution photos enhanced through rotation and scaling, Thaseentaj and Ilango [15] created a deep CNN to identify Anthracnose, Powdery Mildew, and Leaf Blight in mango. Their system achieved 93.34% accuracy. Sharma et al. [16] investigated a CNN-based pipeline for four mango leaf diseases and achieved 90.36% accuracy, while Rao et al. [17] classified grape and mango leaf diseases from a dataset of 8438 photos using a pre-trained AlexNet model, with 99% accuracy for grape leaves and 89% accuracy for mango leaves. The MATLAB-developed system was made available to farmers in real time through the "JIT CROPFIX" Android app.

Uddin et al. [18] proposed a custom YOLOv8-based model named Cauli-Det for the detection and localization of three cauliflower diseases-Bacterial Soft Rot, Downy Mildew, and Black Rot, using a dataset of 656 smartphone-captured images. Their model achieved 93.2% precision, 82.6% recall, and 91.1% mAP, demonstrating strong potential for real-time disease management in the field. Maria et al. [19] and Rajbongshi et al. [20] also investigated CNN and transfer learning models using architectures like InceptionV3, VGG16, and MobileNetV2 for cauliflower disease classification. Maria achieved the highest accuracy of 90.08% with InceptionV3, while Rajbongshi used statistical and feature extraction based on co-occurance with various classifiers, reporting Random Forest as the best performer with 89.00% accuracy, highlighting the growing interest in automated precision monitoring for this high-demand crop.In their evaluation of four DL models for cauliflower disease classification, Pradhan et al. [21] found that EfficientNetB3 outperformed with 84% accuracy, while DenseNet121 with 81%, and ResNet50 with 78%. The findings demonstrate the potential of EfficientNetB3 for sustainable farming methods and early detection.

Jackfruit research is more limited. By combining Random Forest classifiers with K-means segmentation, Habib et al. [22] recognized jackfruit leaf disease with approximately 90% accuracy. Using color transformation, segmentation, and texture features, Arivazhagan et al. [23] proposed a texture-based Support Vector Machine (SVM) method for automatically detecting and classifying leaf diseases in 10 different plant across 13 disease classes. On a dataset of 500 leaves, the accuracy of the method was 94%.

In order to classify grape leaves and diagnose diseases, Kunduracioglu and Pacal [24] looked into the use of 14 CNNs and 17 Vision Transformer models. They made use of the PlantVillage dataset, which contained 4062 photos from four different classes (Black Rot, Leaf Blight, Esca, and Healthy), as well as the Grapevine dataset, which contained 500 photos from five different grape varieties. According to experimental results, a number of models produced exceptional outcomes, with four models most notably

the Swinv2-Base model achieving 100% accuracy. Their research shows how CNN and Transformer based models can be used to accurately identify grape varieties and detect grape diseases, supporting digital crop management and sustainable agriculture. Using 2173 images from a standard dataset, Mane et al. [25] created a customized CNN-based model for detecting potato leaf disease, with an accuracy of 99.6%. For classification, the model made use of structural characteristics like color, intensity, and shape. This strategy proved to be more effective than conventional techniques, providing a dependable means of automated diagnosis of potato disease. Choudhary et al. used the PlantVillage dataset, which contains photos of both healthy and diseased leaves, to create deep learning-based models for identifying diseases in tomato plant leaves. To improve the robustness and generalization of the model, they used data augmentation techniques like zooms, flips, and random rotations. Metrics like accuracy, precision, recall, and F1-score were used to train and assess a number of deep learning models, including AlexNet, DenseNet, LeNet, and two custom CNN architectures. The model with the highest accuracy, DenseNet, was 89%. A custom-designed model came in second with an accuracy of 86.84% and an F1-score of 86% [26]. With an emphasis on the ResNet family, Kunduracioglu [27] assessed deep learning models for tomato leaf disease detection. With 99.85% accuracy and a 99.82% F1-score, the Res2Next50 model produced the best results; Res2Net50d also did well, with 99.78% accuracy. Both models demonstrated the superior ability of sophisticated ResNet architectures for accurate tomato disease classification, outperforming traditional CNNs like VGG16 and DenseNet121. Özüpak et al. used the Kaggle dataset of 4062 photos in five classes to combine MobileNetV2 and Vision Transformer (ViT) in a hybrid stacking model for maize leaf disease classification. With the help of Explainable AI techniques like SHAP, LIME, Grad-CAM and transfer learning, augmentation, and dropout, the model was able to improve generalization and achieve an accuracy of 96.73% [28]. With an accuracy of 99.0%, Aslan and Özüpak used Vision Transformers (ViT) to detect and classify apple leaf diseases early. Their method performed better than previous models, demonstrating the usefulness of ViT in conjunction with sophisticated preprocessing methods for accurate and dependable apple disease diagnosis [29]. Using the Corn Leaf Disease Dataset, Alpsalaz et al. [30] proposed a lightweight and interpretable CNN-based model for classifying maize leaf diseases, achieving 94.97% accuracy with a micro-average AUC of 0.99. The model outperformed benchmark models like ResNet50, MobileNetV2, and EfficientNetB0 and supports real-time mobile deployment with just 1.22 M parameters. For apple leaf disease detection, Kunduracioglu [31] assessed a number of CNN architectures, such as ResNet50, InceptionV4, Xception, DenseNet121, EfficientNetV2-m, and VGG13. Out of all of these, EfficientNetV2-m performed the best, exhibiting superiority over other deep learning models for accurate diagnosis of apple disease with 100% accuracy and F1-score.

Researchers have also explored multi-crop disease recognition. To identify plant leaf diseases, Chohan et al. [32] created a CNN-based deep learning model with 38 classes from the PlantVillage dataset. The model achieved 98.3% testing accuracy using 15% of the dataset and also performed well on real-world images from Sukkur IBA University, while Albattah et al. [33] used the PlantVillage dataset, which included 38 classes, to propose a Custom CenterNet model with a DenseNet-77 backbone for leaf disease detection. Their model outperformed other DL-based and ML-based methods in terms

of precision, recall, and mAP, achieving a state-of-the-art accuracy of 99.98%. However, because of its computational complexity, the model is not yet appropriate for mobile deployment. Khan et al. [34] achieved 99.5% accuracy using a lightweight MobileNetV3-Small model on the PlantVillage dataset (14 crops, 6 disease groups). Deployment on edge devices using the ONNX format was made possible by post-training quantization, which decreased parameters from 1.5 to 0.93 M without sacrificing accuracy. Using BIM, DCGAN, and NST techniques, Pandian et al. [35] suggested a unique 14-layered DCNN trained on a dataset of 147,500 images across 58 plant disease classes and one no-leaf class. In every important metric, the model outperformed transfer learning techniques, achieving 99.97% accuracy. Using a dataset comprising 39 classes of diseased and healthy leaves from 14 plant species, Sharma et al. [36] refined a number of deep learning architectures-VGG16, AlexNet, ResNet18, and MobileNetV2-for the classification of plant leaf diseases. Addressing the shortcomings of small datasets in earlier studies and facilitating early disease detection to lower crop loss were their objectives. As measured by accuracy, F1-score, precision, and recall, MobileNetV2 produced the highest accuracy of 94.4% among the tested models. Khalid et al. [37] used the CNN and MobileNet architectures to detect plant diseases, with 89% and 96% accuracy, respectively. Mohanty et al. [7] achieved 99.35% accuracy by training a deep CNN on 54,306 images from the PlantVillage dataset, which covered 14 crop species and 26 disease classes. Their research proved that deep learning could be used to diagnose diseases with smartphones.

In conclusion, although earlier studies have achieved excellent accuracy in classifying plant diseases under controlled conditions, far fewer have examined generalization using field-captured images showing diverse environmental conditions. Furthermore, there is an opportunity to focus more on Bangladesh's important food and nutritional crops while keeping the models lightweight for resource-limited deployment. Our work expands on this background by putting forth a customized MobileNetV3Large framework trained on a hybrid dataset that includes field sourced images.This approach aims to advance the state of the art for plant disease detection in Bangladesh which will help in future studies to tackle issues like lighting variability, background clutter, and practical deployment.

3 Materials and methods

There are several important steps in the method used in this study. First, images were gathered from public sources and local farms to make a hybrid dataset. After that, the images were preprocessed by resizing, normalizing, and adding more data to make them more varied. We defined and encoded the class labels. The dataset was divided into three parts: training, validation, and testing. Then, a custom MobileNetV3Large model was made, trained, and fine-tuned using the training dataset. Finally, we used different metrics to test the trained model on the testing dataset to see how well it worked. This iterative pipeline makes it possible for continuous improvement and strong evaluation. The overall workflow follows the general deep learning-based approach commonly used for plant disease detection in different studies [38]. Figure 1 shows how the proposed system will work as a whole.

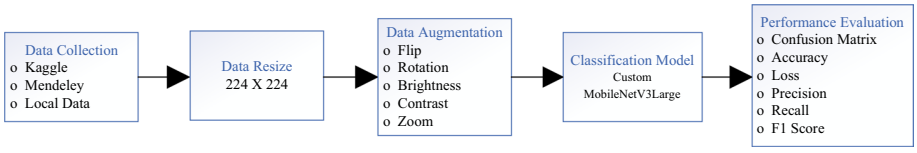


Fig. 1 Overall methodology of the proposed work

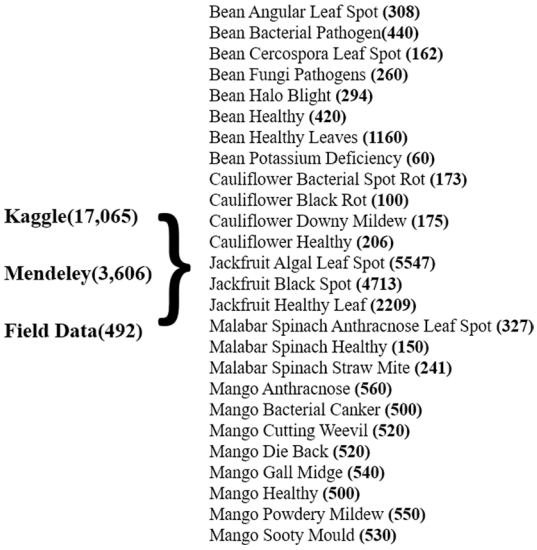


Fig. 2 Class distribution of the collected dataset from different sources

3.1 Data acquisition and dataset splitting

A hybrid dataset was created by merging field-captured photos taken from Bangladeshi farms with photos sourced from public sources such as Mendeley [39, 40] and Kaggle [41–43]. The dataset covers five important crop categories: Bean, Cauliflower, Jackfruit, Malabar Spinach, and Mango, with 26 classes (20 infected, 6 healthy). A total of 21,163 photographs were gathered, providing that a range of backgrounds and lighting conditions typical of real-world situations were covered. To provide a trustworthy and objective assessment of the model, the dataset was split into divisions for training (80%), validation (10%), and testing (10%), and a stratified split was used to ensure proportional representation of all classes across each subset. The multiclass classification framework's automated class mapping was made simpler by the directory-based data structure. Images from public repositories, such as Kaggle and Mendeley, were combined with field-captured photos taken from Bangladeshi farms to create a hybrid dataset. Figure 2 provides a clearer presentation of the class distributions of all 26 categories used in this study, showing how image counts vary among various data sources and plant classes.

3.2 Data preprocessing and augmentation

The size and quality of each gathered image differed, hence a standardized preparation workflow was required. To fit the input size needed by MobileNetV3Large, each image was shrunk to 224 by 224 pixels. In order to align the data with pretrained ImageNet weights, pixel normalization was conducted using the preprocessing function unique to MobileNetV3. Data augmentation samples, illustrated in Fig. 3, that included random zooming, brightness and contrast modifications, random rotations, and random



Fig. 3 Data augmentation samples for the proposed method

horizontal flipping was put into place to further increase dataset diversity and lower the chance of overfitting. In order to expose the model to a greater variety of transformations and increase its resilience to inherent image variability, these augmentations were applied dynamically during training.

3.3 Data labeling and encoding

Class labels, organized within directories that represented each class, were assigned according to crop type and disease category. This method simplified the label encoding procedure by enabling the data pipeline to automatically translate directory names to integer class indices. In order to guarantee annotation quality and reduce mislabeling-both of which are essential for dependable model training-careful labeling validation was carried out, cross-referencing field photos with agricultural references.

3.4 Dataset partitioning

The dataset was split 80:10:10 into training, validation, and test sets after preprocessing and labeling. In addition to offering enough samples for model tuning and final assessment, this partitioning technique guarantees enough data for reliable model training. The test set provided an objective evaluation of the final model performance, the validation set enabled hyperparameter adjustment and early halting, and the training set was utilized to optimize the model weights.

3.5 Model architecture

The proposed model is developed using the existing MobileNetV3Large architecture, chosen for its efficient design and ability to capture complex visual representations. The network was able to apply widely applicable visual features, like edges, textures, and shapes, to the plant disease classification task because the backbone had already been trained on the ImageNet dataset. The original MobileNetV3Large classification head

was eliminated in order to modify the model for this domain. In its place, a dataset-specific classifier comprising 26 classes that covered both diseased and healthy leaf categories was used. The convolutional backbone was first frozen to maintain the generic low-level visual features that ImageNet had already taught it, ensuring stability throughout the transfer learning process. The spatial information from the feature maps was then compressed into a compact representation using a GlobalAveragePooling2D layer, which also lessened the possibility of overfitting brought on by an excessive number of trainable parameters. In order to add non-linearity, two fully connected (Dense) layers containing 256 and 128 neurons were added. Both layers were activated using the ReLU function. To further enhance generalization, dropout regularization was added after each dense layer at rates of 0.3 and 0.2. Lastly, class probabilities corresponding to the plant disease categories were produced using a Dense softmax layer with 26 units. The training was divided into two stages. All layers of the MobileNetV3Large backbone were frozen during the initial transfer learning phase to guarantee that only the newly added classification layers were updated. The deeper layers of the backbone (from index 35 onward) were selectively unfrozen during the fine-tuning stage, while the earlier layers were frozen to preserve stable low-level representations. This made it possible for the deeper layers to adjust to disease characteristics unique to a given domain. The model found a balance between learning task-specific representations and reusing generic features by combining trainable and frozen layers. Images resized to 224×224 pixels were accepted by the model with a batch size of 64 for all training stages in order to standardize the input pipeline. TensorFlow's preprocessing layers were used to create an online augmentation pipeline that reduced overfitting and enhanced generalization. Random rotations up to ±20%, brightness and contrast adjustments within ±20%, random zooming up to ±30%, and random horizontal flipping (probability = 0.5) were among the applied transformations. Before being sent into the network, every image was then normalized using the MobileNetV3 preprocessing function. Overall, the final architecture combines a custom classification head with the effectiveness and lightweight structure of MobileNetV3Large to produce a reliable yet computationally efficient model that is ideal for plant disease detection and deployment on mobile platforms.

In order to serve the objective of field-level plant disease detection on hardware with limited resources, this architectural design strikes a compromise between accuracy and computing efficiency. Figure 4 displays a diagram of this entire design.

A custom convolutional neural network was designed and trained to classify plant disease images into 26 categories. Pixel intensities were normalized to the range [0, 1], and the input images were standardized to a resolution of 224×224 pixels with three RGB channels. In order to capture hierarchical feature representations, the architecture used a sequential design with three convolutional blocks, each of which gradually increased the number of filters. A 2D convolutional layer with 32 3×3 filters and ReLU activation was used in the first block. Batch normalization was then used to stabilize learning, and a 2×2 max pooling operation was used for spatial downsampling. 64 filters with the same setup were used in the second block, which was again followed by batch normalization and pooling. Using 128 filters, batch normalization, pooling, and a dropout rate of 0.3 to lessen overfitting, the third block deepened the network. Following a dropout of 0.5 for more robust regularization, the resultant feature maps were compressed into a one-dimensional vector and run through a fully connected layer of 256 neurons with

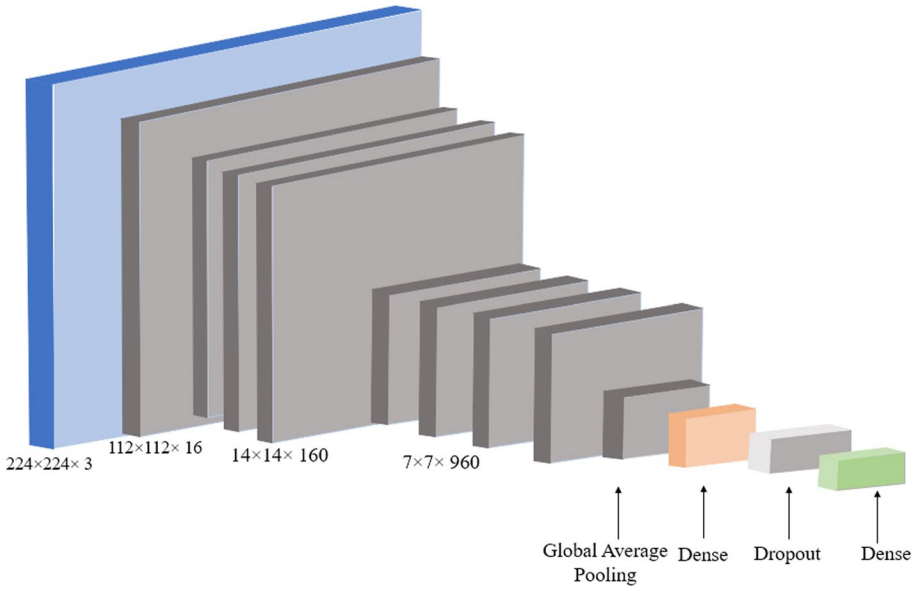


Fig. 4 MobileNetV3Large architect

ReLU activation. Lastly, class probabilities matching the target categories were generated using a dense softmax output layer with 26 units. Accuracy was tracked as the main performance metric while the network was trained using the Adam optimizer with a learning rate of 0.0001 and categorical cross-entropy loss. Model checkpointing and early stopping with ten epochs of patience were used to guarantee robust training, restoring the top performing weights based on validation loss. This setup made the model suitable for identifying fine-grained plant disease categories in the dataset by enabling it to learn both low-level texture and edge features in early layers and higher-level semantic representations in deeper layers. The standard, unaltered implementation of the MobileNetV3Large architecture delivered in TensorFlow Keras, pretrained on ImageNet and modified solely with a task-specific classification head, is referred to in this study as baseline MobileNetV3Large. Similarly, the original pretrained model with no custom modifications is indicated by EfficientNetB1. In order to prevent confusion between the standard implementation and customized version, we specifically use the term baseline to differentiate the unaltered MobileNet from the custom MobileNetV3Large designed in this work.

3.6 Training strategy

The model was trained using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 64. No weight decay was used, and the optimization parameters were set to β_1 = 0.9 and β_2 = 0.999. The sparse categorical cross-entropy loss function was used to deal with the multiclass output. A ReduceLROnPlateau scheduler was used to automatically lower the learning rate when the validation performance plateaued, and early stopping was applied with a three-epoch patience to avoid overfitting. The weights of the top-performing model based on validation loss were also stored using model check pointing. Every training experiment was carried out using a Jupyter Notebook. There were two stages to the training process. All of the MobileNetV3Large backbone's layers were frozen during the first feature extraction phase, making it possible to train

only the recently added classification layers. In order to ensure that the recently added dense layers could learn domain-specific features while preserving stable low-level representations from the pretrained network, this phase lasted for 20 epochs. The deeper layers of the base model were unfrozen for joint optimization in the fine-tuning step that followed, while the top 35 layers were left frozen. The learning rate was lowered to 0.0001 ($0.1\times$ of the initial rate), while the optimizer configuration stayed the same. The same early stopping and learning rate scheduling techniques were used during the next 15 training epochs. This two-phase method helped strike a balance between task-specific learning and feature reuse, enabling the model to successfully adjust to the plant disease dataset without overfitting.

3.7 Computational environment

Every experiment was carried out using the Keras API in TensorFlow 2.18. A personal workstation with an Intel(R) Core(TM) i5-1035G1 CPU running at 1.00 GHz, 8 GB of RAM, and Windows 11 (64-bit) was used for training and evaluation. Since there was no dedicated GPU, the CPU was used for all calculations. This shows that the suggested method can be replicated on common, resource-constrained hardware, even though it lengthened the training period in comparison to GPU-accelerated setups.

4 Results and discussion

This section provides a thorough assessment of the proposed custom MobileNetV-3Large model, comparing it to three reference models: a custom CNN architecture, EfficientNetB1, and a baseline MobileNetV3Large. We evaluated the models on a hybrid dataset with 26 classes of five different plant species. Accuracy and loss curves, confusion matrices, precision-recall curves, Receiver Operating Characteristic-Area Under the Curve (ROC-AUC) curves, and a final comparative accuracy discussion were used for evaluating performance.

4.1 Performance evaluation

The trained custom MobileNetV3Large model was evaluated on the held-out testing dataset using standard classification metrics [44]: accuracy, precision, recall, F1-score, and AUC. These metrics provide a comprehensive view of the systems predictive capacity. For clarity, their definitions are as follows:

Accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

which measures the proportion of correctly classified samples out of the total.

Precision is defined as

$$\text{Precision} = \frac{TP}{TP + FP}$$

which quantifies how many of the positively predicted samples are actually correct.

Recall is defined as

$$\text{Recall} = \frac{TP}{TP + FN}$$

which measures how many of the actual positive samples are captured by the model.

F1-score is defined as

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

which represents the harmonic mean of precision and recall, balancing both. Following the calculation of these metrics for each assessed model, the results are shown in Table 1, which provides a comparison of the precision, recall, and F1-Score values for each model.

4.2 Statistical significance analysis

A paired t-test was used to confirm the models' performance differences' robustness. A statistical technique for figuring out whether the mean difference between two related samples deviates significantly from zero is the paired t-test. To determine whether the improvements were statistically significant and not the result of chance variation, it was applied to the per-class F1-scores of the baseline and suggested models. The proposed model demonstrated statistically significant improvements over the customized CNN (t = 3.23, p = 0.0034), EfficientNetB1 (t = 3.62, p = 0.0013), and baseline MobileNetV3Large (t = 3.62, p = 0.0013) architectures, according to paired t-test analyses performed on the per-class F1-scores. These findings confirm that the observed performance improvements are a consistent and significant improvement in classification ability rather than the result of random variation.

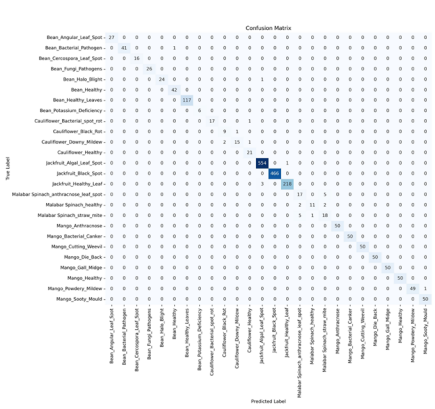
4.3 Confusion matrix

For every model, overall confusion matrices for each of the 26 classes were produced which is shown in Fig. 5. The Custom MobileNetV3 in Fig. 5C is the best-performing model. This conclusion is based on its highly dominant diagonal, which shows an extremely low level of misclassifications with almost all samples correctly classified and almost zero off-diagonal cells. On the other hand, as evidenced by the numerous samples dispersed throughout its off-diagonal cells, the Custom CNN shown in Fig. 5B performs the worst, displaying notable confusion between several classes. Compared to the fine tuned Custom MobileNetV3, the Baseline MobileNetV3 in Fig. 5A and EfficientNetB1 in Fig. 5D models exhibit moderate performance but significantly higher classification errors. The confusion matrices reinforce the superior generalization of the fine-tuned MobileNetV3Large even when tested under diverse real-world scenarios involving background clutter and lighting variation.

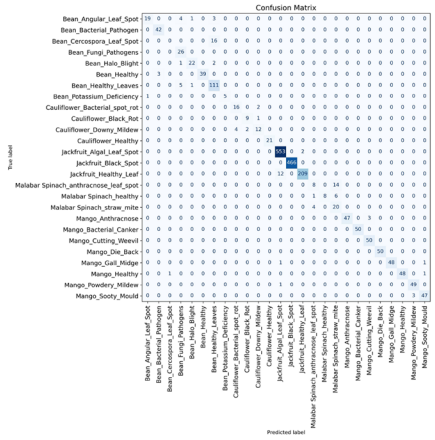
The confusion matrix was constructed for the overall dataset to interpret the distribution of true positives, true negatives, false positives, and false negatives across all 26 classes, providing valuable insight into misclassifications patterns. In addition, the ROC and precision-recall curves were plotted to illustrate the discrimination capability of the

Table 1 Performance comparison of different models

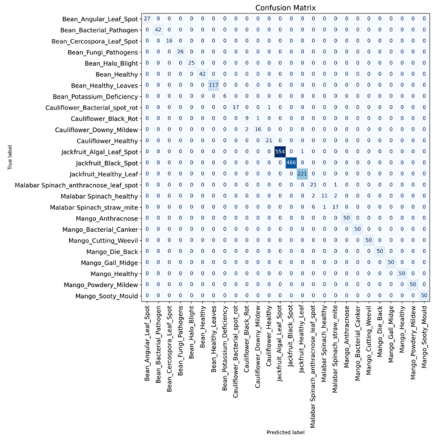
Model	Precision	Recall	F1-score	Accuracy (%)
Custom CNN	0.96	0.96	0.95	95.51
EfficientNetB1	0.98	0.98	0.98	98.30
Baseline MobileNetV3	0.99	0.99	0.99	98.97
Custom MobileNetV3	0.99	0.99	0.99	99.37



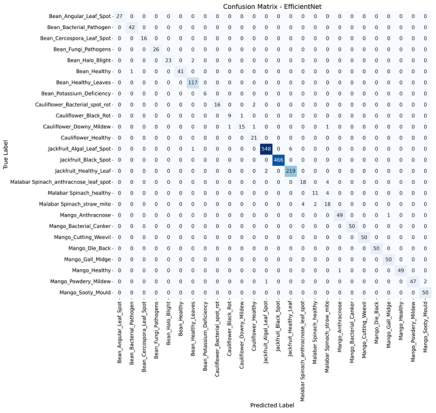
(A) Confusion matrix of Baseline MobilenetV3



(B) Confusion matrix of Custom CNN



(C) Confusion matrix of Custom MobileNetV3



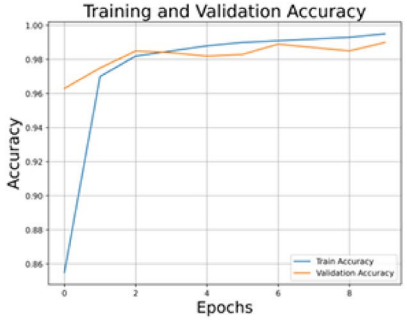
(D) Confusion matrix of EfficientNetB1

Fig. 5 Confusion matrices for four models (**A**–**D**)

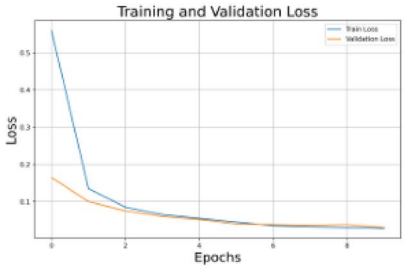
multiclass classifier, which is especially important given the imbalance of some classes in the dataset.

4.4 Accuracy and loss curves

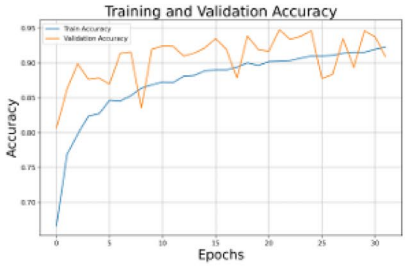
The accuracy and loss curves of the models clearly show their training stability and convergence. As evidenced by continuously increasing training and validation accuracy along with decreasing validation loss over epochs, all models showed efficient learning with little overfitting. The Custom MobileNetV3Large showed the most reliable and effective learning dynamics out of all of them. Its training and validation curves quickly merged and remained stable with few oscillations, as seen in Fig. 6E and F. This behavior shows rapid convergence to low loss and high accuracy, demonstrating learning efficiency and robustness. The Custom CNN, on the other hand, displayed the least amount of stability as shown in Fig. 6C and D. Its accuracy curves varied noticeably, indicating unstable training phases and weaker generalization, while its validation loss was more variable, with spikes in later epochs. Both EfficientNetB1 in Fig. 6G and H and Baseline MobileNetV3 in Fig. 6A and B demonstrated strong performance with smoothly decreasing loss and steadily increasing accuracy. However, the Custom MobileNetV3 continuously outperformed them in terms of synchronization and training dynamics reliability, while their convergence was relatively slower and less stable.



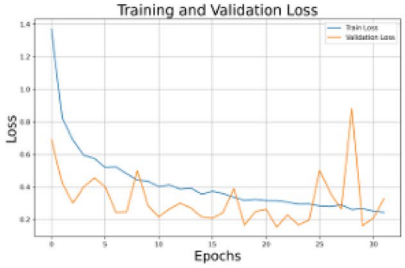
(A) Accuracy curve of Baseline MobileNetV3



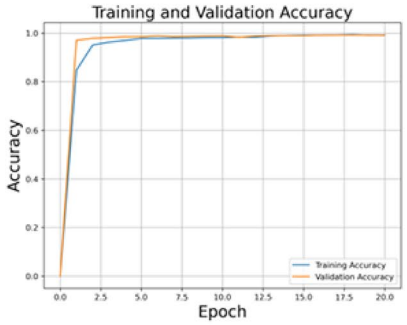
(B) Loss curve of Baseline MobileNetV3



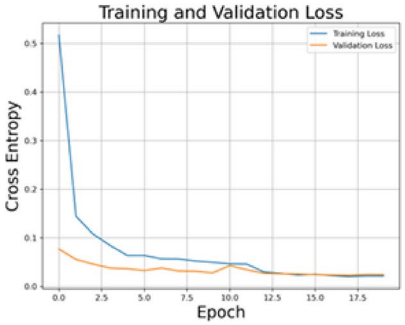
(C) Accuracy curve of Custom CNN



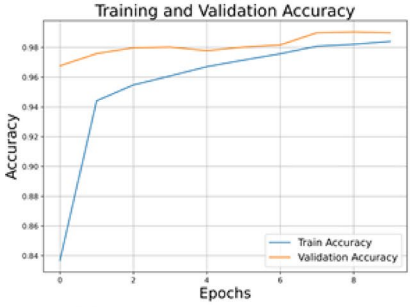
(D) Loss curve of Custom CNN



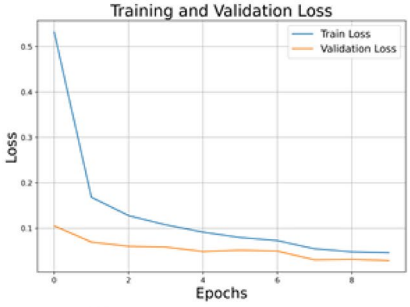
(E) Accuracy curve of Custom MobileNetV3



(F) Loss curve of Custom MobileNetV3



(G) Accuracy curve of EfficientNetB1



(H) Loss curve of EfficientNetB1

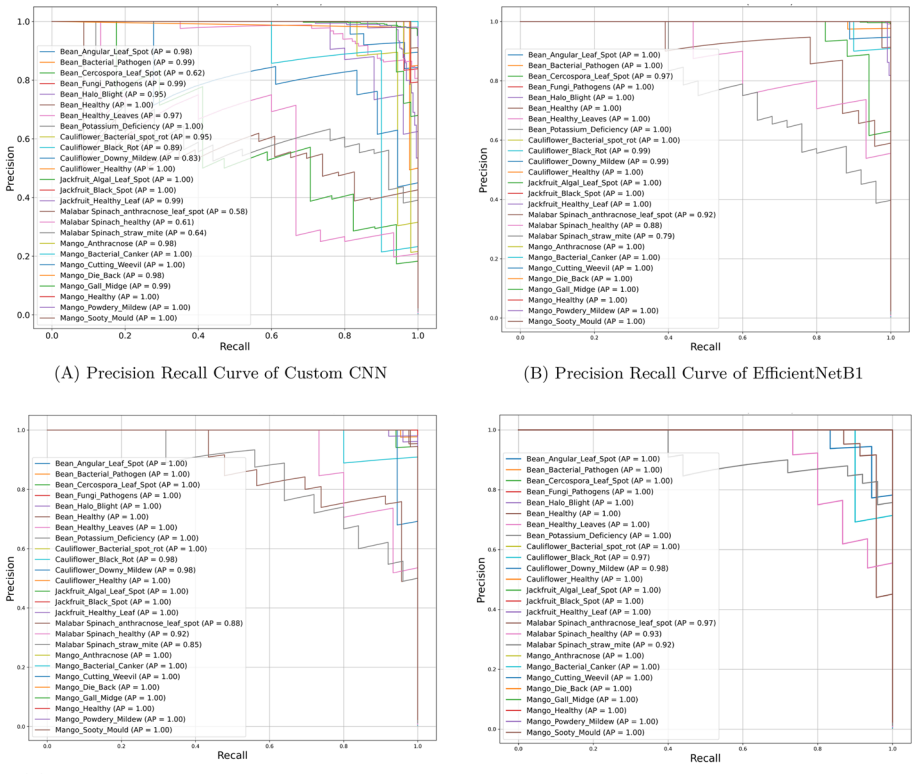
Fig. 6 Accuracy and loss curves for four models: Baseline MobileNetV3 (**A**, **B**), Custom CNN (**C**, **D**), Custom MobileNetV3 (**E**, **F**), and EfficientNetB1 (**G**, **H**)

These figures collectively illustrate that the custom MobileNetV3Large achieved higher stability and less variance in performance compared to the baseline or custom CNN, reflecting the benefits of fine-tuning and well-chosen data augmentation.

4.5 Precision-recall curve

The Precision-Recall (PR) curves of each model are compared in Fig. 7, where Fig. 7A displays the custom CNN, Fig. 7B displays the EfficientNetB1, Fig. 7C displays the baseline MobileNetV3, and Fig. 7D displays the Custom MobileNetV3. The Custom MobileNetV3 outperforms the others in terms of overall performance, continuously maintaining high recall and precision across a variety of classes. This suggests better robustness in managing class imbalance and accurate differentiation between false positives and true positives. Strong PR scores are also attained by Baseline MobileNetV3 and EfficientNetB1, although both exhibit slight precision drops for underrepresented classes. Conversely, the Custom CNN exhibits the lowest performance, with lower recall and precision, especially in visually similar disease categories, underscoring its limited capacity for generalization.

A thorough class-by-class evaluation of the suggested Custom MobileNetV3Large model was carried out in addition to the overall precision–recall analysis. The precision, recall, and F1-score for each disease class are compiled in Table 2, which offers more detailed information on the model's capacity to correctly identify distinct plant disease types.



(A) Precision Recall Curve of Custom CNN

(B) Precision Recall Curve of EfficientNetB1

(C) Precision Recall Curve of Baseline MobileNetV3

(D) Precision Recall Curve of Custom MobileNetV3

Fig. 7 Precision recall curves for four models (**A**–**D**)

Table 2 Class-wise performance metrics of the proposed model

Class	Precision	Recall	F1-score
Bean angular leaf spot	1.00	1.00	1.00
Bean bacterial pathogen	1.00	1.00	1.00
Bean cercospora leaf spot	1.00	1.00	1.00
Bean fungi pathogens	1.00	1.00	1.00
Bean halo blight	1.00	1.00	1.00
Bean healthy	1.00	1.00	1.00
Bean healthy leaves	1.00	1.00	1.00
Bean potassium deficiency	1.00	1.00	1.00
Cauliflower bacterial spot rot	1.00	1.00	1.00
Cauliflower black rot	0.90	0.90	0.90
Cauliflower downy mildew	0.94	0.94	0.94
Cauliflower healthy	1.00	1.00	1.00
Jackfruit algal leaf spot	1.00	1.00	1.00
Jackfruit black spot	1.00	1.00	1.00
Jackfruit healthy leaf	1.00	1.00	1.00
Malabar spinach anthracnose leaf spot	0.70	0.95	0.81
Malabar spinach healthy	1.00	0.73	0.85
Malabar spinach straw mite	0.90	0.75	0.82
Mango anthracnose	1.00	1.00	1.00
Mango bacterial canker	1.00	1.00	1.00
Mango cutting weevil	1.00	1.00	1.00
Mango die back	1.00	1.00	1.00
Mango gall midge	1.00	1.00	1.00
Mango healthy	1.00	1.00	1.00
Mango powdery mildew	1.00	1.00	1.00
Mango sooty mould	1.00	1.00	1.00

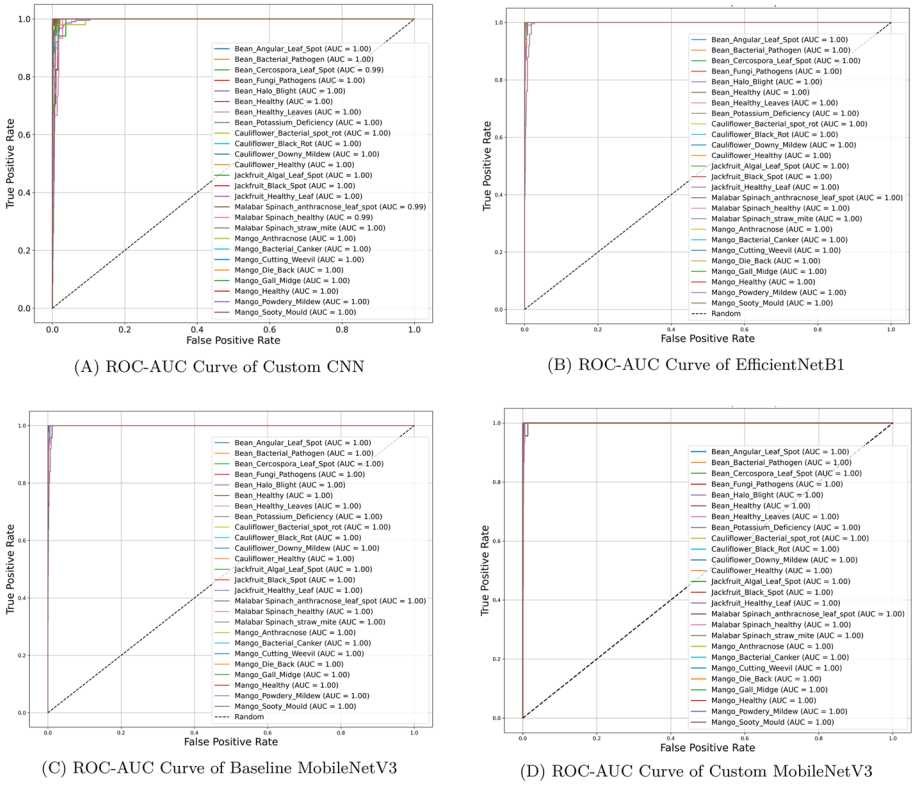
4.6 ROC-AUC curve

Figure 8 shows the ROC curves along with the associated AUC values. With AUC values for almost all 26 classes approaching 1.00, all four models collectively exhibit exceptional discriminative performance. Although one class has a somewhat lower AUC than the others, the Custom CNN model in Fig. 8A exhibits good results. As seen by its steep ROC curve, the EfficientNetB1 in Fig. 8B continuously maintains high true positive rates with few false positives. Similar to this, the Baseline MobileNetV3 in Fig. 8C has curves near the optimal upper left corner and achieves robust classification across classes. Reliable separation is demonstrated across all classes by the Custom MobileNetV3 as shown in Fig. 8D. The results of the Custom MobileNetV3, Baseline MobileNetV3, and EfficientNetB1 are slightly better and more consistent than those of the Custom CNN, demonstrating their greater capacity for generalization.

These findings demonstrate the strong diagnostic capability of the MobileNetV3Large-based system, particularly in multiclass scenarios with subtle differences among disease features.

4.7 Model accuracy comparison

All the four models were then subjected to a comparative performance analysis, as shown in Fig. 9. With a test accuracy of 99.37%, the custom MobileNetV3Large outperformed the baseline MobileNetV3Large (98.97%), EfficientNetB1 (98.3%), and the custom CNN (95.51%). A well-tuned network architecture with fine-tuning layers,



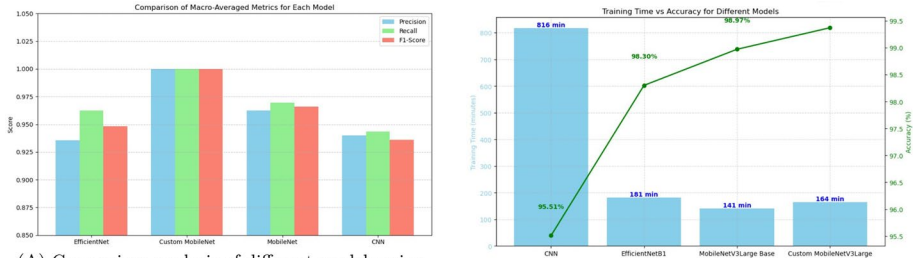
(A) ROC-AUC Curve of Custom CNN

(B) ROC-AUC Curve of EfficientNetB1

(C) ROC-AUC Curve of Baseline MobileNetV3

(D) ROC-AUC Curve of Custom MobileNetV3

Fig. 8 ROC-AUC curves for four models (**A**–**D**)



(A) Comparison analysis of different models using evaluation metrics

(B) Comparison of classifier accuracy versus time

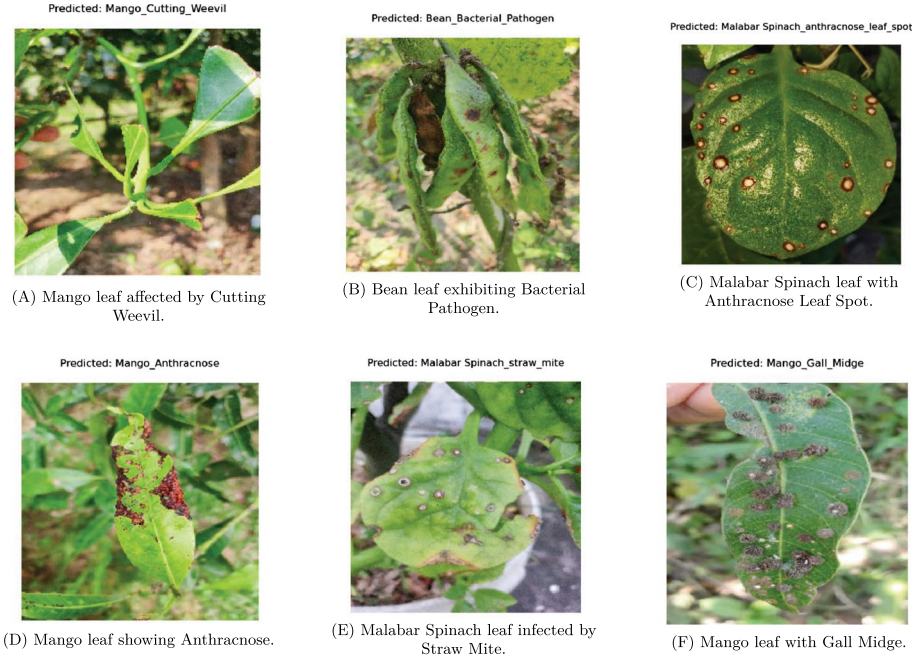
Fig. 9 **A** Shows a comparative analysis of models based on evaluation metrics. **B** illustrates accuracy versus time performance of the classifiers

optimized data augmentation techniques, and the inclusion of field data are all responsible for this improvement.

Overall, the results validate the effectiveness of the proposed system for practical use, confirming its potential for lightweight deployment in smartphones in rural Bangladeshi farming environments.

4.8 Qualitative prediction results

To further validate the effectiveness of the proposed Custom MobileNetV3 model, sample predictions on field-acquired test images are presented in Fig. 10. Numerous plant species and disease categories, such as Mango Cutting Weevil, Bean Bacterial Pathogen, Malabar Spinach Anthracnose Leaf Spot, Mango Anthracnose, Malabar Spinach Straw Mite, and Mango Gall Midge, were successfully classified by the model. These findings



(A) Mango leaf affected by Cutting Weevil.

(B) Bean leaf exhibiting Bacterial Pathogen.

(C) Malabar Spinach leaf with Anthracnose Leaf Spot.

(D) Mango leaf showing Anthracnose.

(E) Malabar Spinach leaf infected by Straw Mite.

(F) Mango leaf with Gall Midge.

Fig. 10 Sample qualitative predictions of plant diseases using the proposed Custom MobileNetV3 model

show that the model consistently performs well across a variety of visual patterns and disease types, even when field conditions include variations like background noise, overlapping symptoms, and uncontrolled lighting. The precise forecasts demonstrate the suggested framework's resilience and versatility for practical uses. Through class-specific predictions, the model also offers an interpretable decision support tool that can help farmers and agricultural experts quickly and accurately identify plant diseases. The field-captured photos only made up a small portion of the dataset which remains a limitation of this work. However, the incorporation of field data has strengthened the testing process's robustness. Also, the model has demonstrated consistent performance on these real-world samples. In future works, more real time field data can be incorporated to further improve the model. A comparative study with previous studies is presented in Table 3. It can be seen that our dataset is more diverse considering both public and field data. Also, this study has considered more classes of plant diseases than the previous works. The result of the designed model is satisfactory as well.

5 Conclusion

A deep learning-based method for accurately classifying plant leaf diseases that impact several economically and nutritionally significant crops in Bangladesh is presented in this paper. Using a hybrid dataset that included 26 classes of healthy and diseased leaf images from both public repositories and field-captured samples, a customized MobileNetV3Large model was created and trained. Because of its lightweight and computationally efficient design, the model can be used in environments with limited resources. Numerous tests showed that the suggested model performed better than baseline architectures, such as the MobileNetV3Large, EfficientNetB1, and a customized CNN. A paired t-test was performed on the per-class F1-scores to make sure these improvements weren't random. The results showed statistically significant improvements over baseline MobileNetV3Large ($t = 3.62$, $p = 0.0013$), customized CNN ($t = 3.23$, $p = 0.0034$), and

Table 3 Comparative analysis of existing studies and the proposed model

Reference	Method	Dataset	Classes	Accuracy
Haque et al. [14]	CNN	Field collected	2 Classes of Malabar Nightshade	96.77%
Elfatimi et al. [11]	MobileNetV2	Publicly available	3 Bean classes	92.97%
Thaseentaj et al. [15]	DCNN	Field collected	3 Mango diseasest	93.34%
Chohan et al. [32]	CNN	Publicly available and Field collected	4 Classes for apple, grape and corn, 3 classes for potato, 2 classes for peach, cherry and bell pepper, strawberry, 10 classes for tomato, 1 class for blueberry, raspberry, soybean, squash and orange	98.3%
Pradhan et al. [21]	EfficientNetB3	Field collected	4 Cauliflower classes	98%
Khalid et al. [37]	MobileNetV2	Publicly available	20 distinct classes of crops	96%
Our proposed model	Custom MobileNetV3	Publicly available and field collected	8 Classes of bean, 4 classes of cauliflower, 3 classes of jackfruit, 3 classes of malabar spinach, 8 classes of mango	99.37%

EfficientNetB1 (t = 3.62, p = 0.0013). This demonstrates the superior performance of the suggested model and its dependability and consistency. The modified model performed well on other evaluation metrics like precision, recall, F1-score, ROC-AUC, and confusion matrix analysis, and it obtained a superior test accuracy of 99.37%. These findings support the model's efficacy and generalizability in a range of real-world scenarios, such as changes in lighting, background complexity, and noise in field images. Although the results are encouraging, there are still some restrictions. Because different crops and disease types are not always available at the same time throughout the year, it was not possible to gather field data for every disease class due to seasonal variability. Despite several logistical challenges, every effort was made to collect field data. Furthermore, in real-world deployment, class imbalance may be introduced because not all diseases in the dataset are equally prevalent in field conditions. The fact that the mobile application itself has not yet been implemented, despite the suggested model being optimized for mobile and edge deployment, is another drawback. In order to guarantee real-time, accessible disease detection for farming communities, future work will concentrate on adding more varied, seasonally distributed field images to the dataset, implementing sophisticated segmentation techniques for accurate lesion detection, and implementing the model via a mobile or web-based platform.

Author contributions
Jabin Rahaman: conceptualization, formal analysis, methodology, visualization, writing—original draft. Priyanti Paul: conceptualization, methodology, visualization, writingoriginal draft. Aditta Chowdhury: conceptualization, visualization, writing—original draft. Muhammad Quamruzzaman: supervision, visualization, writing—review.

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Data availability
Data is open-access and cited within the manuscript.

Declarations

Ethics approval and consent to participate
This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for publication
Not applicable.

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The authors declare that they have no conflict of interest.

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